

Simple Linear Regression: Fitting

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MATH 240

Week 4, Lecture #2 — Ch. 7

Updates and Reminders

- Homework #3 is due **Friday** at 11:59pm
- Lab #2 during **Monday's** class
 - ▶ We will finish the simple linear regression slides during Friday's class

- Simple linear regression
 - ▶ Interpretations
 - ▶ Strength
 - ▶ Use of categorical predictors
- Performing a fit
 - 1 Exploratory data analysis (EDA)
 - 2 Fitting
 - 3 Diagnostics
 - 4 Interpretations of coefficients
 - 5 Predictions

Objectives

- Use R to fit a least-squares line to a bivariate relationship
- Diagnose various plots to verify conditions for a least-squares fit

Interpretations

- **Intercept:** The estimated value of y when $x = 0$
- **Slope:** The estimated **average** change in y for every one unit change in x
 - ▶ When slope is positive, then y increases as x increases
 - ▶ When slope is negative, then y decreases as x increases

Examples: Interpretations

Say we have the following data and least squares line on the displacement (x) and MPG (y) of various cars from the late-20th century:

	1	2	3	4	5	6	7	8	9	10
x	160.0	160.0	108.0	258.0	360.0	225.0	360.0	146.7	140.8	167.6
y	21.0	21.0	22.8	21.4	18.7	18.1	14.3	24.4	22.8	19.2

$$\hat{y} = 25.5638 - 0.0249x$$

- **Intercept:** 25.5638 represents the MPG of a car with zero displacement
- **Slope:** For every one unit increase in displacement, then there will be a 0.0249 decrease in MPG on average

Extrapolation

- **Extrapolation:** applying a model estimate to values outside of the realm of the original data
- Outside of the domain of our data, we do not know how the data will behave
- Extrapolation assumes that the linear relationship holds in areas of the domain where it has not been analyzed

Extrapolation Example

Say we have the following least squares line on the displacement (x) and MPG (y) of various cars from the late-20th century:

$$\hat{y} = 25.5638 - 0.0249x$$

Say some auto manufacturer wanted to create a car with a displacement of 1200. If we tried to predict this car's MPG using the least squares line found from the 10 cars, we would get the following:

$$\begin{aligned}\hat{y} &= 25.5638 - 0.0249(1200) \\ &= -4.3162\end{aligned}$$

Having a negative MPG does not make sense in this case. Hence, we need to be careful about extrapolation.

Strength of Model: Use of R^2

- We want to estimate the strength of the linear relationship between two variables
- While the correlation coefficient R can be used, it is more common to use the R-squared value R^2
- R^2 : describes amount of variation in response explained by the least squares line
 - ▶ Goes from 0 to 1
 - ▶ $R^2 = 1 \implies y_i - \hat{y}_i = 0$ for all i

Use of R^2

We can calculate R^2 in the following ways:

$$R^2 = 1 - \frac{SSE}{SST} \quad (1)$$

$$= 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (2)$$

$$= \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

$$= \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right) \quad (4)$$

- Can be used to describe the goodness of fit for your model (1, 2)
- Can be used to show how much variation in the response variable can be explained by the model (3)
- Is the squared correlation coefficient between x and y (4)
- The larger the R^2 , the better the fit

Use of Categorical Predictors

- Categorical predictors can be used in a simple linear regression
- Use of indicator variable to specify if coefficient is associated with that group
- Examples: male/female, blue/brown/green, yes/no
- For each variable with k groups, we need $k - 1$ indicator variables
 - ▶ e.g. x_1 : 1 if someone said yes, 0 otherwise
 - ▶ e.g. x_1, x_2 :
 - ★ 1 for x_1 and 0 for x_2 if someone has blue eyes
 - ★ 0 for x_1 and 1 for x_2 if someone has brown eyes
 - ★ 0 for x_1 and 0 for x_2 if someone has green eyes

Example

Say we have the following data and least squares line on the transmission (x) and MPG (y) of various cars from the late-20th century:

	1	2	3	4	5	6	7	8	9	10
x	1	1	1	0	0	0	0	0	0	0
y	21.0	21.0	22.8	21.4	18.7	18.1	14.3	24.4	22.8	19.2

$$\hat{y} = 19.843 + 1.757x$$

- 19.843 represents the MPG of a car with an automatic transmission on average (e.g. $x = 0$)
- 21.6 represents the MPG of a car with a manual transmission on average (e.g. $x = 1$)

Tutorial: mtcars Dataset

- Consists of fuel consumption and 10 aspects of automobile design and performance for 32 cars from 1973-74
- Data was extracted from the 1974 Motor Trend US magazine
- Number of observations $n = 32$; number of features $p = 11$

Tutorial: mtcars Dataset

```
mtcars  
head(mtcars)
```

	mpg	cyl	disp	hp	drat	wt
Mazda RX4	21.00	6.00	160.00	110.00	3.90	2.62
Mazda RX4 Wag	21.00	6.00	160.00	110.00	3.90	2.88
Datsun 710	22.80	4.00	108.00	93.00	3.85	2.32
Hornet 4 Drive	21.40	6.00	258.00	110.00	3.08	3.21
Hornet Sportabout	18.70	8.00	360.00	175.00	3.15	3.44
Valiant	18.10	6.00	225.00	105.00	2.76	3.46

	qsec	vs	am	gear	carb
Mazda RX4	16.46	0.00	1.00	4.00	4.00
Mazda RX4 Wag	17.02	0.00	1.00	4.00	4.00
Datsun 710	18.61	1.00	1.00	4.00	1.00
Hornet 4 Drive	19.44	1.00	0.00	3.00	1.00
Hornet Sportabout	17.02	0.00	0.00	3.00	2.00
Valiant	20.22	1.00	0.00	3.00	1.00

Steps to Keep in Mind

Suppose we cared about how the MPG affected the displacement of a vehicle (using this dataset). We will want to do the following tasks when performing linear regression:

- 1 Exploratory data analysis (EDA)
- 2 **Linear regression fitting**
- 3 **Diagnostics**
- 4 **Interpretation** and hypothesis testing of coefficients
- 5 Prediction (with confidence and prediction intervals)

1. Exploratory Data Analysis (EDA)

- Identifying **general patterns** in the data
- Helps you get familiar with the data and come up with potential hypotheses for linear regression
- Completed before linear regression
- Can be in the form of:
 - ▶ Plots and graphs (e.g. scatter plots, histograms)
 - ▶ Summary tables (e.g. five-number summaries)

1. Exploratory Data Analysis (EDA)

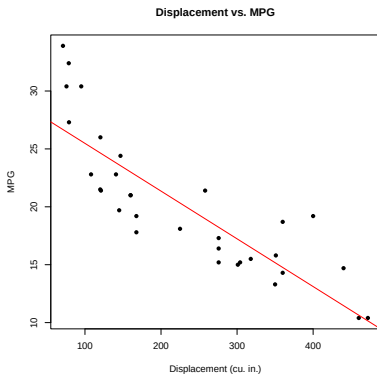
Effective exploratory data analysis involves effective communication. Keep these principles in mind for creating data visualizations:

- Keep it simple (e.g. no extraneous features; should be bare-bones)
- Use color/aesthetics to draw attention
- Make labels easy to read
- Tell a story
- Order matters
- Pick a purpose

Tutorial: EDA with the mtcars Dataset

We can check the relationship between MPG and displacement:

- `plot(mtcars$disp, mtcars$mpg, xlab = "Displacement (cu. in.)", ylab = "MPG", main = "Displacement vs. MPG", pch = 16)`
- `abline(lm(mtcars$mpg ~ mtcars$disp), col = "red")`



2. Linear Regression Fitting

- Will need to fit a linear model between the two variables
- Can analyze conditions after fitting the model
- Aids us in future steps, such as inference and prediction

Main function to use in R:

- `lm(response ~ covariate)`

Note that we want to assign this model to a variable, such as `mod`.

Tutorial: Linear Regression Fitting with the mtcars Dataset

To fit the model, we can use one of the following commands (second one is preferred):

- `mod <- lm(mtcars$mpg ~ mtcars$disp)`
- `mod <- lm(mpg ~ disp, data = mtcars)`

We can then check the summary of this model with the following code:

- `summary(mod)`

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	29.5999	1.2297	24.07	0.0000
disp	-0.0412	0.0047	-8.75	0.0000

Tutorial: Linear Regression Fitting with the mtcars Dataset

- `summary(mod)`

Call:

```
lm(formula = mtcars$mpg ~ mtcars$disp)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.8922	-2.2022	-0.9631	1.6272	7.2305

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	29.599855	1.229720	24.070	< 2e-16 ***
mtcars\$disp	-0.041215	0.004712	-8.747	9.38e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.251 on 30 degrees of freedom

Multiple R-squared: 0.7183, Adjusted R-squared: 0.709

F-statistic: 76.51 on 1 and 30 DF, p-value: 9.38e-10

Activity: Linear Regression Fitting with the mtcars Dataset

Activity: For a pair of two continuous variables in the last activity, please fit a linear model. Use the `summary()` function to identify the β coefficients and interpret them.

3. Diagnostics

Recall the four conditions:

- ① **Linearity**: data shows a linear trend
- ② **Nearly normal residuals**: residuals should follow approximately a normal distribution
- ③ **Constant variability**: variability around regression line should remain constant throughout all x
- ④ **Independent observations**: should not have some underlying structure (e.g. time series data)

We can verify these conditions through diagnostic plots that R has.

3. Diagnostics

We will analyze four different types of plots:

- ① **Residuals vs. Fitted plot:** used to verify **linearity** of relationship
 - ▶ Straight, flat line through points → satisfied
- ② **Normal quantile-quantile (QQ) plot:** used to check if we have **normal residuals**
 - ▶ Points, for the most part, follow 45 degree diagonal line → satisfied
- ③ **Scale-location plot:** used to check homoscedasticity (e.g. **equal variance** among residuals)
 - ▶ Points hover around horizontal line with equally spread points → satisfied
- ④ **Residuals vs. Leverage plot:** used to identify **influential points** (i.e. extreme values that will influence regression model)
 - ▶ Want as much of a horizontal line through the points as possible
 - ▶ Points labeled as influential, typically in corners of the plot

In summary: **follow the dashed line and it's satisfied!**

3. Diagnostics

To get these four plots, we can use the following commands:

- `plot(mod)`

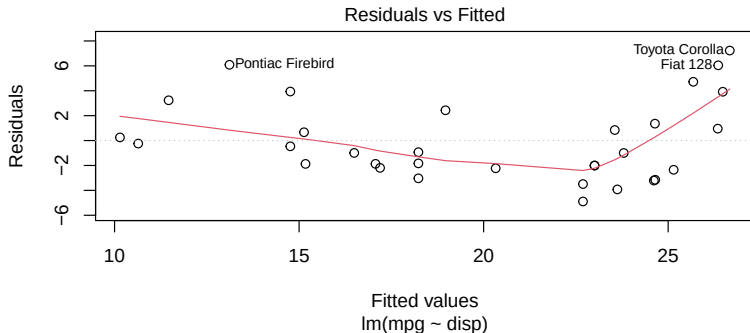
You will then need to hit 'Return' in the console to cycle through the plots. If you want all four plots in one figure, we can use the following commands:

- `par(mfrow = c(2, 2))`
- `plot(mod)`

Tutorial: Diagnostics with the mtcars Dataset

We can look at the Residuals vs. Fitted plot first:

- `plot(mod, 1)`

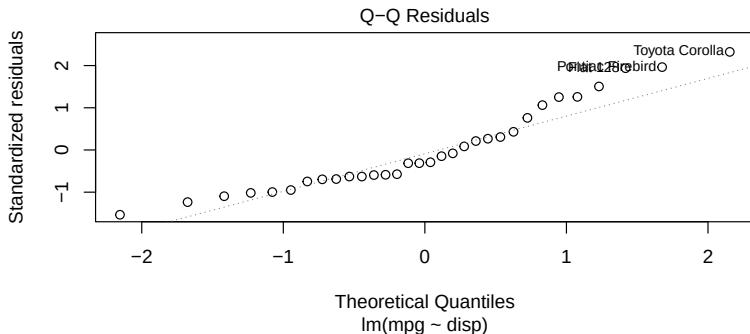


- Some concerns about linearity – residuals are not scattered consistently
- See higher slope in the red line for higher fitted values

Tutorial: Diagnostics with the mtcars Dataset

We can look at the QQ plot next:

- `plot(mod, 2)`

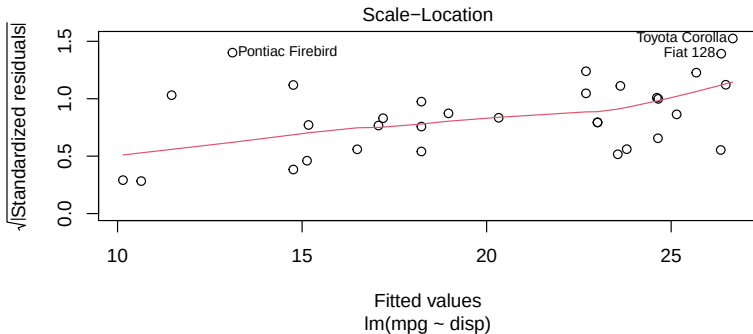


- Mostly normal residuals, but we have some extreme residuals that trail away from line

Tutorial: Diagnostics with the mtcars Dataset

We can look at the scale-location plot first:

- `plot(mod, 3)`

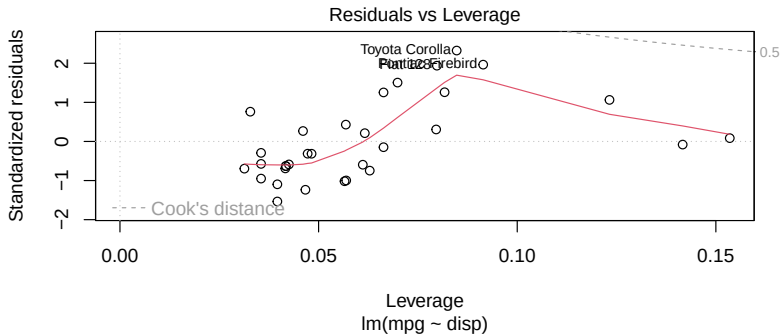


- Points are spread out consistently as we go left to right, but...
- ...the standard residuals increase with higher fitted values

Tutorial: Diagnostics with the mtcars Dataset

We can look at the Residuals vs. Leverage plot next:

- `plot(mod, 5)`

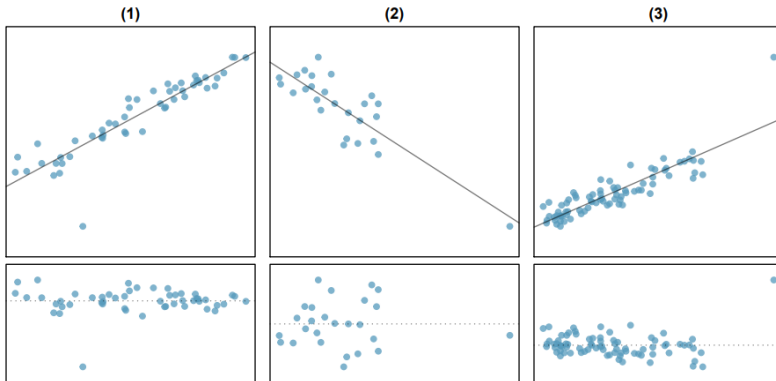


- Points do not surpass Cook's distance
- A few influential points as indicated by apex of red line

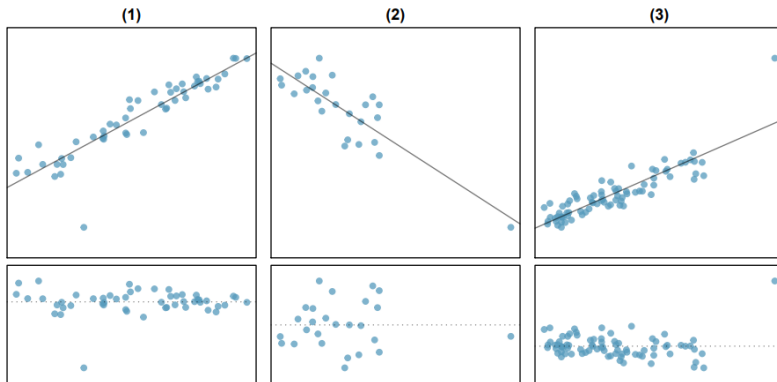
Sidenote: Influential Points

- **Influential point:** data points that affect the estimated coefficients of the least-squares line
 - ▶ When removed, the revised least-squares line will better fit the data
 - ▶ These points are generally far away from the least-squares line

Example: Understanding Influential Points

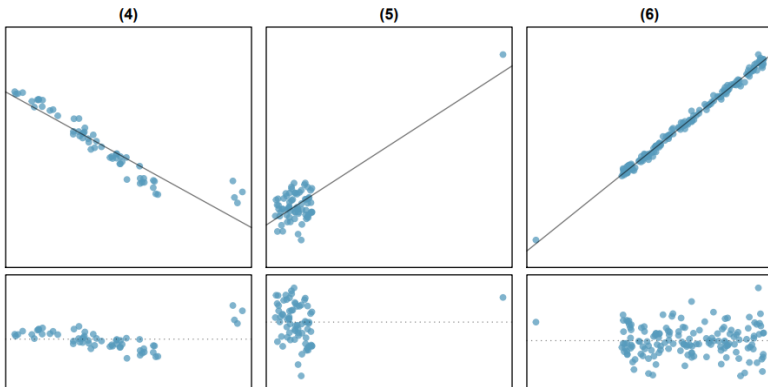


Example: Understanding Influential Points

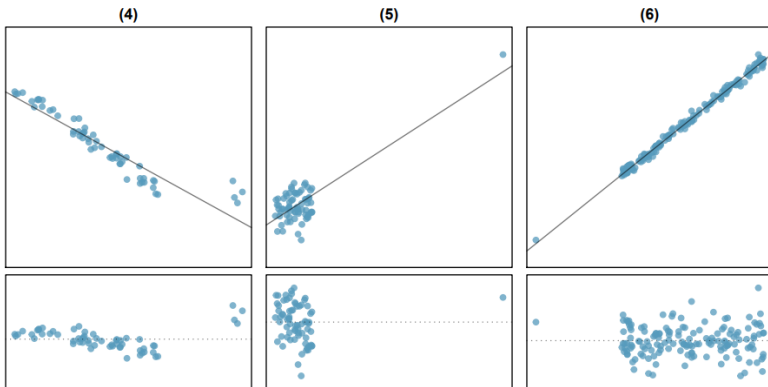


- ① One outlier – slightly influences the line
- ② One outlier – very close to the regression line, not very influential
- ③ One outlier – appears to pull the regression line up

Example: Understanding Influential Points



Example: Understanding Influential Points



- ④ A small cloud of 4 outliers, pulling the regression line up
- ⑤ One outlier, not very influential
- ⑥ One outlier, not very influential

Activity: Diagnostics with the mtcars Dataset

Activity: Comb through the diagnostic plots of the model you created from the previous activity. Make sure to detail what you see in the four plots, as well as justify if the conditions are satisfied:

- **Linearity**
- **Nearly normal residuals**
 - ▶ Outliers?
 - ▶ Influential points?
- **Constant variability**
- **Independent observations**